Katharina **Deckenbach**

PhD student · AI Safety

About me

Highly motivated and curious PhD student with a hard-working attitude. Excited about the potential of AI and driven by the goal of mitigating its risks to ensure future AI systems can be as beneficial as possible.

Contact

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kaethy

katharina-deckenbach

nBGXSEAAAAJ

Languages

German — Native

English — C1

Spanish — C1

Swedish — B1

French — A1

Programming

Python

R

Matlab

Java

Julia

Other Skills

Uncertainty Estimation · Git · PyTorch · Weights&Biases · Hugging Face · Presentation · Communication

Education

03/26 - present **PhD student @ COMPASS Group**
ELLIS Institute Tübingen — Tübingen, Germany
Topic: Evaluation Awareness, supervised by Sahar Abdelnabi

08/23 - 06/25 **M.Sc. Machine Learning**
KTH Royal Institute of Technology — Stockholm, Sweden
Thesis: *"Improving the Laplace Posterior Approximation via Sharpness-Aware Minimization"*.

09/21 - 03/22 **Erasmus Semester**
Universidad Pompeu Fabra — Barcelona, Spain

10/19 - 03/23 **B.Sc. Cognitive Science**
University of Tübingen — Tübingen, Germany
Thesis: *"Are People Confident When Artificial Neural Networks Are Confident? "*.

Experience

11/25 - 02/26 **Research Intern**
Max Planck Institute for Intelligent Systems — Tübingen, Germany
- Research Internship with Sahar Abdelnabi's COMPASS group

07/25 - 10/25 **Research Engineer**
KTH Royal Institute of Technology — Stockholm, Sweden
- Continued my thesis work on improving posterior approximation methods
- Trained and evaluated models (e.g., ResNet, ViT, BERT) in PyTorch, using Laplace Approximation and SWAG

2025 **Participant**
ML4Good and ARBOx — Strasbourg, France and Oxford, UK
- Two 10-day in-person bootcamps to up-skill in AI safety
- Topics covered training ResNet and GPT-2 from scratch, RLHF, reasoning models, mechanistic interpretability, adversarial attacks, frontier model evaluations, steering vectors

04/22 - 07/22 **Teaching Assistant**
University of Tübingen — Tübingen, Germany
- Held weekly tutorial session of "Cognitive Architectures" (topics included: RL, evolutionary algorithms, embodied cognition, perception)

Scholarship

12/20 - 03/22 **Deutschlandstipendium**
Public-private German Scholarship

Engagement

12/25 - present **Co-Organizer AI Safety Tübingen**

01/24 - 06/25 **Co-Organizer Stockholm AI Safety**

Publications

2026 **K. Deckenbach**, H. Puerto, J. Geiping, and S. Abdelnabi, "Models That Know How Evaluations Are Designed Score Safer", in *Advances in Neural Information Processing Systems (NeurIPS) 40*, 2026.

2025 E. Banzuzi, D. J. Khodaverdian, and **K. Deckenbach**, "A Reproducibility Study of Decoupling Feature Extraction and Classification Layers for Calibrated Neural Networks", *Transactions on Machine Learning Research June-2025*, 2025.